

Table 1: Platforms and toolkits for IoT and edge computing

Software	URL	Usage
Node-RED	nodered.org	Flow based programming environment
Contiki	contiki-os.org	Microcontrollers, IPv6, IPv4, protothreads, low resource, game consoles
FlowHub	flowhub.io	Flow based IoT programming
NoFloJS	noflojs.org	JavaScript based flow programming
Netron	github.com/lutzroeder/netron	Dynamic visualisation
PyFlow	wonderworks-software.github.io/PyFlow	Visual scripting
Yet another robot platform (YARP)	yarp.it	Robotic programming
OROCOS	orocos.org	Robotic programming and machine control
OpenIoT	openiot.eu	Sensing as a service (S2aaS)
Zetta	zettajs.org	WebSocket programming, real-time on TCP, reactive programming, low overhead scenarios
DSA	iot-dsa.org	Real-time interfacing, inter-device communication, programming on multiple layers
IoTivity	iotivity.org	Constrained application protocol (CoAP), IoT programming
CupCarbon	cupcarbon.com	Smart city, SCI-WSN simulation, visualisation, 2D and 3D Open-StreetMap, MQTT programming, sensor programming
KAA	kaaproject.org	Data analytics, real-time applications, dynamic communications and updates

Node-RED can be installed on multiple platforms including open source hardware devices, IoT devices, the cloud or on dedicated servers. Its cloud deployment can be done on IBM Cloud, Microsoft Azure, SenseTecnica FRED and Amazon Web Services. To work with IoT, fog and edge scenarios, it can be installed on Raspberry Pi, BeagleBone Black, or Arduino. These devices support the interfacing of IoT sensors for multiple applications.

To run Node-RED on a dedicated system, the `node-red` command is executed to start the server. A local host based IP address with port number is generated, on which the programming for IoT, edge computing, etc, is done.

After starting the server, there are thousands of packages that can be interfaced for multiple applications. To create an IoT or edge scenario, packages related to Wi-Fi, Bluetooth, MQTT,

CoAP, weather forecasting, etc, can be installed from the NPM repository.

After the required packages are installed, the flow diagram is created

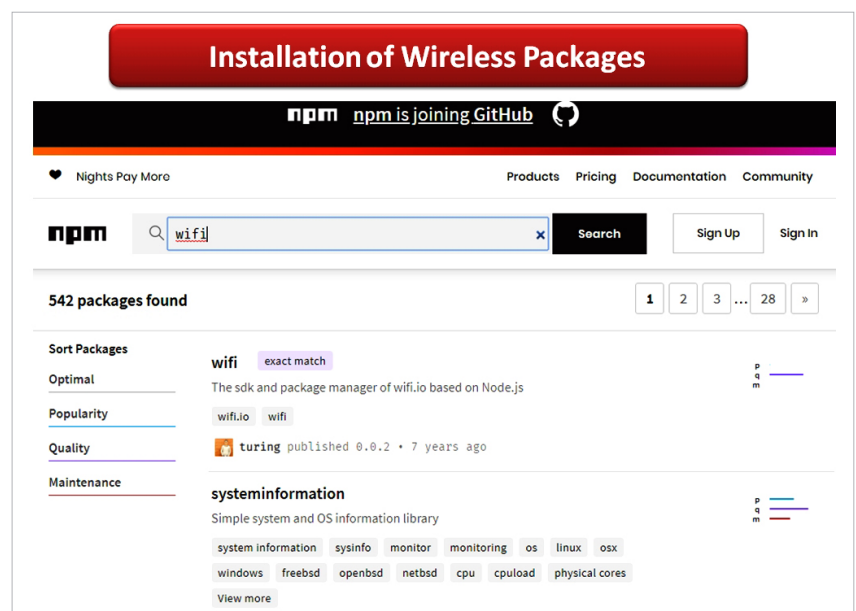


Figure 5: Searching a Wi-Fi package from the NPM repository